RAM CHARAN VANGA
RA2511027010164
Sub ject : DSA

Unit 3, S4, SLO2 - Stack Applications - Infix to Postfix Conversion 1

Write an algorithm for Conversion of Infix Expression to Postfix using Stack.

Answer: Algorithm InfixToPostfix(exp):
Step 1: Push '(' onto Stack, and add ')' to the end of the infix expression exp.
Step 2: Scan the infix expression exp from left to right until the Stack is empty.
Step 3: For each scanned character C:
a. If C is an operand (alphanumeric), append C to the Postfix output string.
b. If C is '(', push C onto the Stack.
c. If C is an operator (+, -, *, /, ^, %):
i. While top of Stack has an operator of higher or equal precedence (or right-associative rules apply):
Pop the operator from Stack and append to Postfix output.
ii. Push C onto the Stack.
d. If C is ')':
i. Repeatedly pop from Stack and append to Postfix output until a '(' is encountered.
ii. Pop and discard the '(' from the Stack.
Step 4: Return Postfix output expression.
Step 5: Exit.